

Logic gates

Computers use electrical signals not only to understand numbers and letters but also to make decisions using devices called “logic gates”. There are four main types of logic gates: “AND”, “NOT”, “OR”, and “EXCLUSIVE OR”.

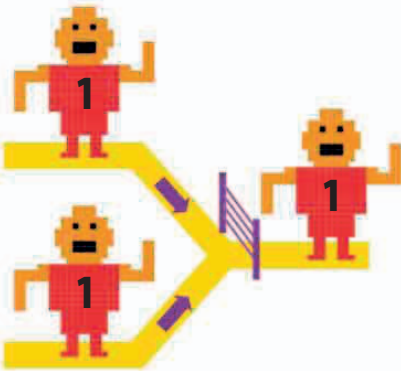
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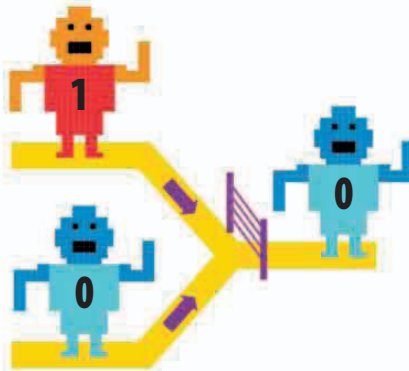
◀ **182–183** Binary and bases

AND gate

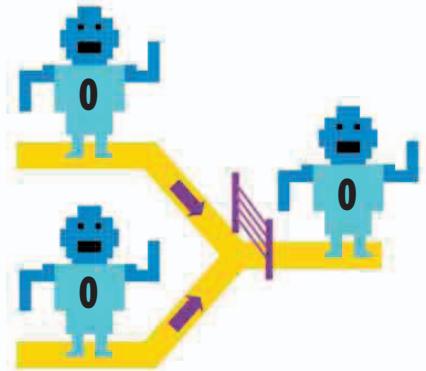
Gates use one or more input signals to produce an output signal, based on a simple rule. AND gates switch their output signal “on” (1) only when both input signals are “on” (1 and 1).



△ **Inputs 1 and 1 = output 1**
Both input signals are “on”, so the AND gate produces an “on” output signal.



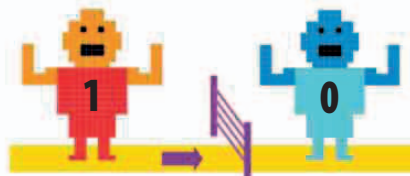
△ **Inputs 1 and 0 = output 0**
If one input is “on” but the other is “off”, the output signal is “off”.



△ **Inputs 0 and 0 = output 0**
An AND gate produces an “off” output signal if both input signals are “off”.

NOT gate

These gates “flip” any input to its opposite. “On” input becomes “off” output, and “off” input turns to “on” output. NOT gates are also known as “inverters”.



△ **Input 1 = output 0**
The NOT gate flips an “on” input to an “off” output, and vice versa.



REAL WORLD

George Boole (1815–64)

George Boole was an English mathematician whose work made logic gates possible. He worked out a system to solve logic problems. This kind of math, which deals in values that can only be true or false (positive or negative), is known as “Boolean logic” in his honor.